

TECHNICAL REQUIREMENTS & SYSTEM ARCHITECTURE PROPOSAL

Enterprise B2B Psychometric & Technical
Assessment Platform

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1. Executive Summary

This proposal outlines the technical architecture and implementation strategy for a standalone, enterprise-grade B2B Unified Psychometric and Technical Assessment Platform. Designed specifically for corporate hiring teams and recruiters, the system eliminates traditional evaluation bottlenecks such as manual test link distribution, candidate tracking friction, and delayed grading. This platform automates the end-to-end assessment lifecycle:

- **For Corporate Clients:** White-labeled, customized test packages linked to job descriptions (JDs).
- **For Candidates:** Frictionless, access-code-based assessment sessions with real-time progress autosaving.
- **For Administrators & HR:** Instant AI-driven scoring summaries and downloadable PDF/HTML reports.

1.1 End-to-End Operational Lifecycle Workflow

To ensure complete operational clarity across stakeholders, the platform executes a streamlined 6-step lifecycle from initial package creation to final candidate reporting:

STEP 1: Admin Package & Access Code Generation Platform Admin creates the assessment package, attaches the Job Description (JD), and generates a unique Access Code (e.g., CORP-2026-X).

STEP 2: Dispatch to Corporate HR Admin shares the Platform Access Link (`` assesspulse.com/login ``) along with the Access Code and Candidate Guidelines with Corporate HR.

STEP 3: Corporate HR Candidate Distribution Corporate HR distributes the Platform Link, Access Code, and Candidate Guidelines directly to their applicant pool via their internal channels.

STEP 4: Candidate Access & Identity Capture Candidate accesses the link, enters the Access Code, and submits BOTH:

1. Full Name (Mandatory human-readable identifier for HR reports)
2. Email Address (Primary unique system key for tracking & verification)

STEP 5: Assessment Execution & Real-Time Autosave Candidate completes the test while answers are continuously saved in the background via debounced API requests (saving every 1.5–5s).

STEP 6: Asynchronous AI Evaluation & HR Report Delivery Upon submission, Gemini 3.1 Flash evaluates responses against the JD, compiles scores, and generates downloadable PDF/HTML reports for HR.

2. Background & Problem Statement

2.1 The Problem

Evaluating job applicants using traditional psychometric testing methods introduces significant operational overhead for corporate HR teams:

1. **Manual Administration Overhead:** HR teams must manually generate access credentials, track candidate progress across external tools, and compile evaluation scores.
2. **Delayed Turnaround:** Multi-day delays occur between test completion and when hiring managers receive actionable report insights.
3. **Data Privacy Restrictions:** Corporate HR departments often hesitate to upload or expose candidate email lists to third-party tools due to data governance regulations.
4. **Static & Uncontextualized Grading:** Standard assessment models score candidates on static metrics rather than evaluating candidate responses against specific role requirements (Job Descriptions).

2.2 The Solution

The platform provides a secure, white-labeled B2B SaaS application. Corporate HR administrators create an umbrella account, generate time-bounded batch access codes, and distribute front-facing assessment links internally. When a candidate completes a test, an asynchronous AI pipeline grades their responses directly against the job description and generates a print-ready report in seconds.

3. Project Objectives (SMART)

- **Specific:** Build a multi-tenant B2B platform with dedicated Admin management views ("Corporates" and "Assessments" library), a candidate portal, an AI scoring engine, and automated PDF report generation.
- **Measurable:** Achieve sub-second response times on candidate autosaves (≤ 200 ms) and generate complete AI PDF reports in under 8 seconds.
- **Achievable:** Utilize a modern, highly scalable technical stack (React, FastAPI, Supabase/PostgreSQL, Gemini 3.1 Flash).
- **Relevant:** Solve primary B2B recruitment bottlenecks for corporate hiring managers and HR personnel.
- **Time-Bound:** Complete platform MVP delivery, system integration, and production cutover within a strict, highly focused 2-week rapid deployment schedule.

4. Scope of Work

4.1 In-Scope Functionality

- **White-Label Corporate Branding:** Neutral UI design customizable per enterprise client.
- **Admin Management Panel:** Dedicated management interfaces for "Corporates" (client management) and 'Assessment Library' (psychometric and technical test package creation).
- **Candidate Identity Capture (Dual Verification):** Candidate entry requires both **Full Name** and **Email Address** alongside the Access Code.
- **Primary Unique Key:** Email Address serves as the system-wide unique identifier for session data and HR tracking.
- **Human Identity Fallback:** Capturing the candidate's Full Name upfront resolves the "unmatched email" risk—if a candidate uses a personal email address (e.g., john.personal@gmail.com) instead of the corporate email registered on HR's records, HR can still identify the candidate in their tracking dashboard via their verified Full Name.
- **Real-Time Response Autosave:** React client-side debounced state persistence (saving continuously every 1.5–5 seconds).
- **AI Evaluation Engine:** Asynchronous Google Gemini 3.1 Flash API integration using structured JSON schema enforcement to grade candidates against Job Descriptions.

- **Resilient Report Generation:** Native WeasyPrint PDF compiler featuring an automatic static HTML fallback mechanism.
- **HR Progress Tracking Dashboard:** Dedicated tracking URL (/track/{uuid_token}) providing completion metrics, and CSV export capabilities.
- **Automated Reminder Engine:** A background task scheduler that automatically dispatches email nudges to pending candidates exactly 24 hours before their deadline.
- **Embedded Candidate Support:** In-assessment technical issue modal (/support/submit) capturing session metadata for immediate engineering alerts.

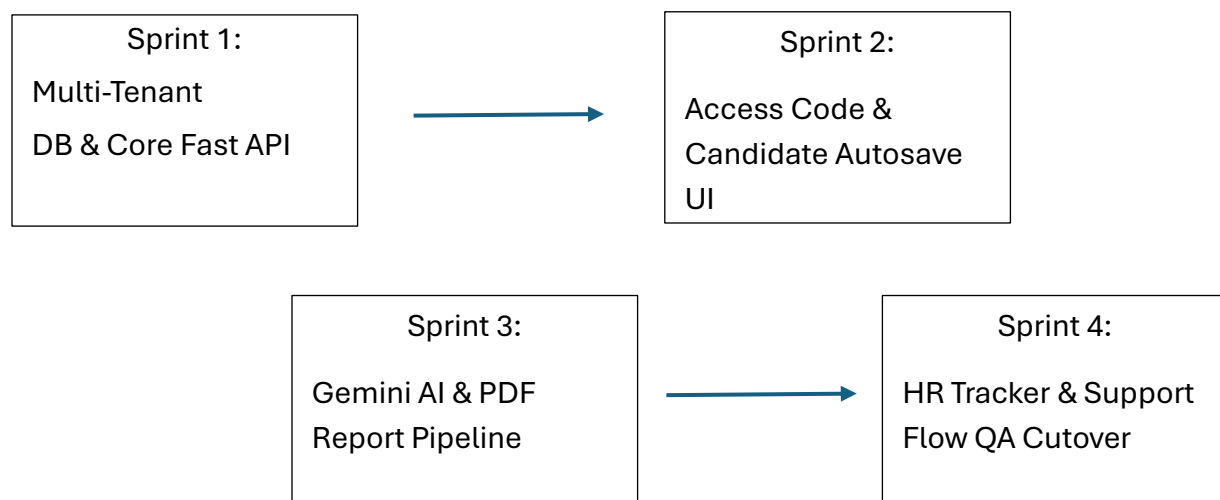
4.2 Out-of-Scope (Future Iterations)

- Native mobile apps (iOS/Android) — web application is fully responsive across mobile and desktop browsers.
- Direct ATS (Applicant Tracking System) API webhooks in Version 1.0.

5. Software Process Model & QA

5.1 Chosen Model: Agile / Iterative Scrum

The project follows a Rapid Application Development (RAD) model, executed as a single, intensive 2-week sprint. This approach enables rapid iteration and continuous verification across four distinct development phases.



5.2 Quality Assurance & Testing Strategy

- **Unit Testing:** Pytest test suite validating API route logic, Mod-10 access code generators, and CSV exporters.
- **Integration Testing:** Verification of candidate authentication, response upserts, and support ticket creation endpoints.
- **E2E Smoke Testing:** End-to-end user workflow verification: *Admin creates package → Code generated → Candidate completes test → AI evaluates response → PDF report generated.*

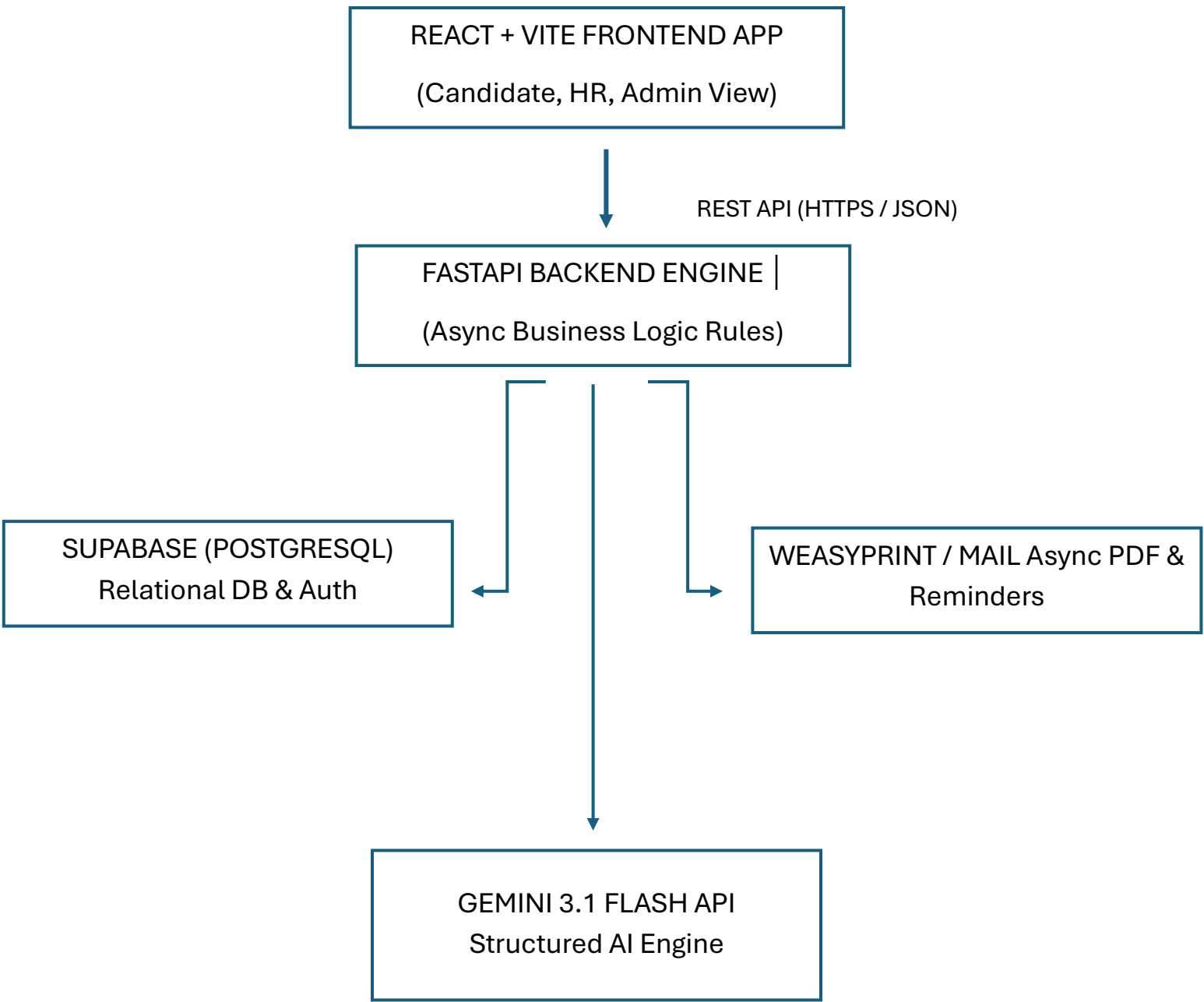
6. System Architecture & Tech Stack

6.1 Technology Stack Overview

Layer	Selected Technology	Purpose
Frontend	React + Vite + TypeScript	High-performance single-page candidate and administrative portals.
Styling	Tailwind CSS + Radix/Shadcn	Modern, white-label corporate UI components.
Backend API	FastAPI (Python 3.11+)	Asynchronous REST APIs handling business logic and worker queues.
Database	Supabase (PostgreSQL)	Managed PostgreSQL with Row Level Security (RLS) for multi-tenant data isolation.
AI Scoring	Gemini 3.1 Flash API	Structured JSON evaluation of candidate responses against Job Descriptions.
PDF Engine	WeasyPrint (HTML → PDF)	Renders print-ready PDF reports locally with static HTML fallback.

Email Service	SendGrid / AWS SES	Transactional email engine for candidate reminders and technical alerts.
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6.2 High-Level Architecture Diagram



7. System UI/UX Structure

7.1 Admin Portal Layout

- **Dashboard Tab:** Key performance indicators (active corporate accounts, completion rates, active candidate sessions).
- **Corporates Tab:** Umbrella view to onboard B2B enterprise clients and configure custom branding parameters (colors, logos, and client contact details). *(Future Recommendation: Tiered usage quotas and account caps).*
- **Assessment Library Tab:** Master assessment library allowing admins to compose packages by bundling standard psychometric modules, technical assessments, and operational tests
- **Package Builder:** Guided interface to structure test sections, configure question types (Likert, MCQ, Open-Ended), and attach Job Descriptions. Allows administrators to mix psychometric and technical assessments, enforced with a strict system limit of up to 6 assessments per package

7.2 Candidate Portal Layout

- Login Screen: Access code + candidate identity interface
- Access Code Input: Pre-filled when accessed via direct candidate links (?code=CORP-YYYY-XXXXX-C).
- Full Name Input (Mandatory): Captures candidate's legal/formal name to ensure corporate HR can immediately identify applicants even if they use an alternate personal email.
- Email Address Input (Mandatory): Serves as the primary unique key for candidate session data and tracking.
 - Guideline Onboarding: Pre-test overview outlining instructions, time limits, autosave confirmation, and support contact options.
 - Single-Question Test Runner: Focused test runner interface dynamically rendering input components:
- Likert Scale: 1–5 agreement selection scale.
- Multiple Choice (MCQ): Radio-button single selection lists.
- Open-Ended: Text area with real-time background autosaving.
 - Embedded Technical Support Modal: Floating issue form allowing candidates to report technical difficulties without losing progress.

7.3 HR Tracking Dashboard (/track/{uuid_token})

- **Live Completion Status:** Real-time summary metrics showing Invited, In-Progress, and Completed candidate sessions.
- **Data Privacy Protection:** Masked PII view hiding raw access codes and partial candidate emails.
- **Action Bar:** "Export CSV" control and to dispatch automated email nudges to pending candidates.

8. Technical Specifications

8.1 Database Schema Entity-Relationship Structure

The application utilizes six core relational tables in PostgreSQL:

- **corporates:** B2B tenant umbrella storing company profiles, custom branding assets (logos, primary colors), and enterprise metadata.
- **assessments:** Master library of reusable test modules, psychometric question definitions, technical assessments, and operational test banks.
- **packages:** Corporate test configurations containing assigned assessment modules, generated access codes, and Job Description (JD) text for AI grading.
- **candidates:** Participant session records storing full_name (human-readable identity for HR cross-referencing), email (primary system key), and a due_date (TIMESTAMP type to capture the exact date and time for T-24h automated reminders), all bound to a specific package ID.
- **candidate_progress:** Assessment status tracking (NOT_STARTED, IN_PROGRESS, COMPLETED), trait scores, and AI evaluation output payloads.
- **candidate_responses:** Individual question answers persisted incrementally via real-time upsert logic (UNIQUE: candidate_id, test_id, question_id).

8.2 AI Integration Engine (Gemini 3.1 Flash)

- **Prompt Strategy:** The candidate's raw answers and the package's Job Description are submitted asynchronously to Gemini 3.1 Flash.
- **Output Format:** Enforced JSON Schema returning qualitative trait scores (1–10), key behavioral strengths, areas of caution, and recommended interview follow-up questions.
- **Execution:** Non-blocking background worker process executing upon test submission.

8.3 Check-Digit Access Code Formula

Access codes are constructed with an integrated checksum digit using the Mod-10 algorithm over character ASCII values:

$$C = \left(\sum_{i=1}^n \text{ord}(\text{char}_i) \right) \bmod 10$$

This structure allows the API to validate code integrity instantly before executing database queries.

9. Risk Assessment & Mitigation Matrix

Risk Scenario	Impact	Likelihood	Mitigation Strategy
WeasyPrint System Dependency Failure	High	Medium	Catch ImportError/OSError and fall back to writing a fully styled static HTML report file.
Database Lock During Autosave Spikes	High	Low	Deploy on PostgreSQL (Supabase) to handle parallel writes and high connection concurrency.
Gemini API Rate Limiting	Medium	Medium	Implement background retry queues with exponential backoff and non-blocking worker threads.

Candidate Network Interruption	High	Medium	Background debounced autosave hook (useAutosave) continuously persists answers every 1.5–5 seconds.
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10. Budget & Cost Estimation

Infrastructure Item	Service Provider	Tier / Usage Model	Estimated Monthly Cost
Backend API Host	Render / Railway	Web Service (0.5 vCPU, 512MB RAM)	\$7.00 / month
Database & Storage	Supabase	Free Tier (Up to 500MB DB) → Pro	\$0.00 – \$25.00 / month
Frontend Hosting	Vercel / Netlify / Cloudflare	Hobby / Pro Tier	\$0.00 / mo
AI Evaluation Engine	Google Gemini API	gemini-3.1-flash Pay-per-token	~\$0.00015 / test (~\$1.50 per 10k tests)
Transactional Email	SendGrid / AWS SES	Developer / Free Tier	\$0.00 / mo
Total Estimated Cost		Staging & Production Pilot	~\$7.00 – \$32.00 / mo

11. Timeline & Milestone Deliverables

MILESTONE 1: Multi-Tenant Foundation (Days 1–3)

Setup Supabase PostgreSQL schema and multi-tenant RLS policies.

Configure FastAPI backend with corporate tenant model relationships.

Deliverable: Functional multi-tenant API framework.

MILESTONE 2: Admin Expansion & Candidate Portals (Days 4–7)

- Build "Corporates" and "Assessment Library" tabs in React, including the 6-assessment package limit logic.
- Implement Candidate runner with debounced autosave and support form.
- *Deliverable:* Working end-to-end testing interface for psychometric and technical modules.

MILESTONE 3: Gemini AI Pipeline & Report Engine (Days 8–10)

- Integrate Gemini 3.1 Flash API with Job Description contextual prompting.
- Configure WeasyPrint async worker pipeline with HTML report fallback.
- Deploy background worker task scheduler (e.g., Celery) for the T-24h automated candidate

MILESTONE 4: HR Tracking, QA Auditing & Cutover (Days 11–14)

- Deploy public HR Progress Tracker (/track/{uuid_token}) with CSV export.
- Conduct end-to-end integration tests and execute production deployment.
- *Deliverable:* Production-ready white-labeled assessment platform.